Comparison Between KNN and SVM on the Saturn Dataset

1. Introduction

This study evaluates and compares the performance of two supervised classification algorithms: **K-Nearest Neighbors (KNN)** and **Support Vector Machines (SVM)**.

The evaluation is performed on the basis of the “Saturn” dataset.

Two validation approaches were applied:

- **Hold-Out Validation** (proportions **80%-20%**)
- **10-Fold Cross Validation**

The purpose of this “double” validation is to ensure a reliable estimate of the model’s generalization performance.

2. Dataset

The dataset we use is called “***Saturn***”. It is a binary classification dataset with 540 istances (275 class = -1 and 255 class = +1) and 8 numeric input features.

The implementation, carried out in R, uses the following libraries:

- **caret**: used for data partitioning, model training and cross-validation.
- **e1071**: used for SVM implementation.
- **klaR**: used for additional classification utilities.
- **dplyr**: used for data manipulation and preprocessing.

3. K-Nearest Neighbors (KNN)

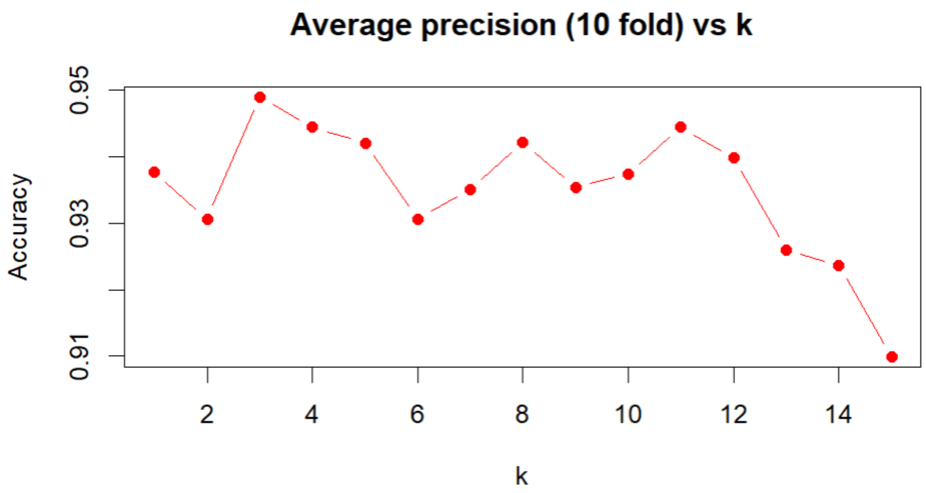
3.1 Preprocessing

The Saturn dataset is divided into two parts: the **Training set (80%)** and the **Test set (20%)**. The former is used for training and validation, the latter is used finally to verify the model’s actual performance (that is the performance on “never-seen” data).

Since KNN is “distance-based”, the data is normalized using the **MIN-MAX method**.

The best value of the **k** parameter is then found using **Internal Cross-Validation**.

The result is shown in the following graph. The best value of k is the point with the highest accuracy (k=3).



3.2 Hold-out and K-fold Cross Validation

They are applied to the Training set which is further divided into **Training set (80%)** and **Validation set (20%)**.

The results of the Hold-out and K-fold Cross validation are briefly described in the following tables.

Table 1

HOLD-OUT VALIDATION	VALUE (%)	ACCURACY (%)
EMPIRICAL RISK	1.45%	98.55%
GENERALIZATION RISK	8.14%	91.86%
FULL EMPIRICAL RISK	2.31%	97.69%
FULL GENERALIZATION RISK	2.78%	97.22%

Table 2

K-FOLD CROSS VALIDATION	VALUE (%)	ACCURACY (%)
EMPIRICAL RISK	2.39%	97.61%
GENERALIZATION RISK	6.03%	93.97%
FULL EMPIRICAL RISK	2.31%	97.69%
FULL GENERALIZATION RISK	2.78%	97.22%

This results show low errors across all stages and negligible difference between training and testing performance (no overfitting is detected). The model shows stability and a very good generalization.

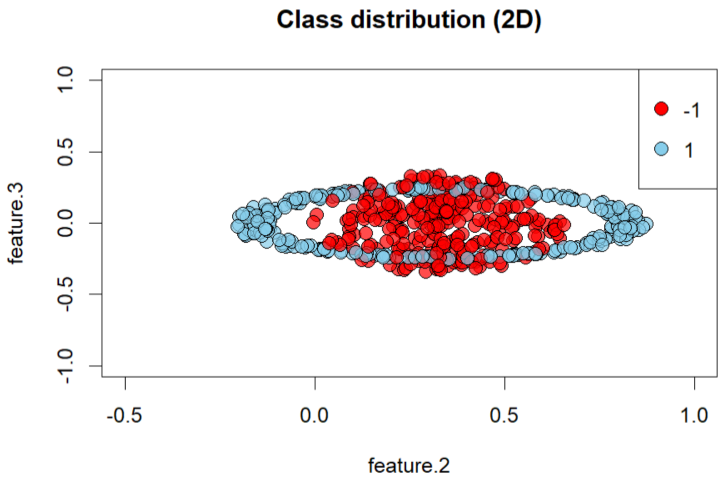
Values that appear larger than others depend on the size of the data set. For example, the **8.14%** and **6.03%** generalization risk values may seem slightly higher compared to the others. However, this is due to the size of the validation set and test set, respectively (recall that the data set has 540 instances).

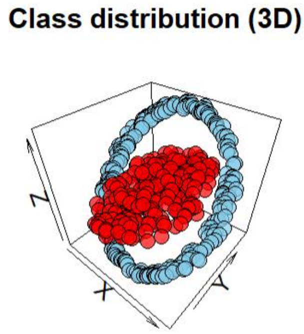
4. Support Vector Machine (SVM)

4.1 Linear separability

If we were working with linearly separable data, then the Hard Margin SVM could be a good classification method. The first analysis on the Training set (kernel equal to ***"linear"***), however, shows a very high empirical error (**33.33%**) and support vectors (**97.96%**).

This is the sign that we are working with **non-linearly separable data** (as shown in the following graphs). So the linear SVM is not good.





The mapping could be a good solution.

4.2 Mapping

The idea is to move from a space where the istances are not linearly separable to a space where, hopefully, they are.

To do this, we reapply the SVM function by changing the kernel to ***"radial"***. This applies the typical **Radial Basis Function (RBF)** mapping.

We can see that the results improve. In fact, we move to an empirical error of **18.98%** and a lower percentage of support vectors (**60.93%**).

Table 3

LINEAR SVM	VALUE (%)
EMPIRICAL RISK	33.33%
SV	97.96%

RADIAL SVM	VALUE (%)
EMPIRICAL RISK	18.98%
SV	60.93%

4.3 Hyperparameter Tuning via K-Fold cross validation

The ***svm* function** of the library "e1071" has default values for the **C** (regularization) and **γ** (kernel) hyperparameters.

The next step is to modify these parameters to find the optimal combination, thus finding a compromise that minimizes the empirical error as much as possible and with a number of support vectors that is not a large percentage of our dataset (**Bias-Variance tradeoff**).

C and **γ** are identified through K-fold cross validation. The results are shown in the following table.

Table 4

KERNEL	C	GAMMA	EMP.ERR.	GEN.ERR.	SVs(%)
Radial	100	0.1	0	0	20.397

4.4 Model Refinement using Optimal Parameters

Using the parameters indicated in the previous paragraph, the **Radial Kernel SVM model** is retrained on the entire training set and predictions are made on the same training set and on the test set (**"never-seen" data**).

In both cases, the model obtained zero error, both in empirical error (training) and generalization error (test).

Table 5

SVM ON TRAINING SET

PRED / ACTUAL	-1	+1
-1	220	0
+1	0	212

FULL.EMP.ERR. (%)	0
SVs(%)	19.91

SVM ON TEST SET

PRED / ACTUAL	-1	+1
-1	55	0
+1	0	53

FULL.GEN.ERR. (%)	0
SVs(%)	19.91

4.5 Final Results

This result indicates that the dataset is completely separable in the space transformed by the RBF kernel, and that the chosen parameter combination allows for perfect distinction between classes.

Regarding the SV points, approximately 1 in 5 points is a support vector, so the model is using only the **20%** of the data to define the decision frontier, while **80%** of the points are already well separable and are not needed to determine the boundary.

We have axcellent equilibrium, a simple and stable model. The dataset is well separable in the new RBF space, has low complexity and it generalizes very well.

5. Conclusion

Both KNN and SVM achieved very low classification errors, with the RBF SVM even reaching **0%** test error.

This result indicates that the “Saturn” dataset is highly separable in the transformed feature space.

The SVM with radial kernel can model the circular decision boundaries more accurately, while KNN, being a local nearest-neighbor method, may incur small errors near class boundaries.

Overll, both models show excellent performance, with the SVM showing a slight advantage in its ability to represent the non linear nature of the dataset.

6. References

Course slides and laboratory from *Machine Learning*, Unimib, 2025.

R packages: *caret*, *class* and *e1071* (used for data partitioning, KNN and SVM implementations)

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